

SILVERDERM CREAM

Contains silver sulfadiazine 1.0% w/w.

Excipients: Carbomer 940, Triethanolamine, Propylene glycol, Cetyl alcohol, Liquid paraffin, Polysorbate 80, Lanolin anhydrous, Chlorhexidine gluconate sol. 20%, Zalibra (colour perfume), Purified water.

Description

A white cream.

Indications

It is indicated as an adjunct for the prevention and treatment of wound sepsis in patients with second and third-degree burns.

Recommended Dosage

The cream should be applied once to twice daily to a thickness of approximately 1/16 inch on the burn areas. It should be continued until satisfactory healing has occurred or until the burn site is ready for grafting.

Route of administration

To be applied topically.
For external use only.

Contraindications

As sulfonamides are known to cause kernicterus, **Silverderm Cream** should not be used at, or near-term pregnancy, on premature infants or on newborn infants during first months of life. **Silverderm Cream** is also contraindicated in patients known to be hypersensitive to silver sulfadiazine or to any of the excipients listed.

Precautions / Warnings

Fatalities associated with the administration of sulfonamides have occurred due to severe reactions, including Steven-Johnson syndrome, toxic epidermal necrolysis and other reactions. **Silverderm Cream** should be discontinued at the first appearance of skin rash or any sign of adverse reaction.

Transient leukopenia has occurred although its association with application of **Silverderm Cream** has not been confirmed. Nevertheless, regular blood counts are advisable in patients on long-term treatment.

Patients should be watched carefully for sensitivity, especially if there are known reactions to sulfonamides.

Local reactions have been reported in patients treated with silver sulfadiazine; the separation of the eschar may be delayed and fungal invasion of the wound may occur.

In patients with extensive burns, serum sulfonamide concentrations and renal function should be

monitored and urine examined for sulfonamide crystals. Absorption of propylene glycol contained in the cream can affect serum osmolality which can interfere with some laboratory tests.

Silverderm cream contains lanolin dan cetyl alcohol. These excipients may cause local skin reactions (e.g. contact dermatitis).

Silverderm cream contains chlorhexidine. Chlorhexidine is known to induce hypersensitivity, including **generalised** allergic reactions and anaphylactic shock. The prevalence of chlorhexidine hypersensitivity is unknown, but available literature suggests this is likely to be very rare. **Silverderm cream** should not be administered to anyone with a possible history of an allergic reaction to chlorhexidine.

If any signs or symptoms of hypersensitivity reaction such as itching, skin rash, redness, swelling, breathing difficulties, light headedness, and rapid heart rate develop, immediately stop using the product. Appropriate therapeutic counter-measures must be instituted as clinically indicated.

The use of **Silverderm cream** in some cases of glucose-6-phosphate dehydrogenase-deficient patients may be hazardous as haemolysis may occur.

During treatment of burns over a large body area, significant amounts of silver sulfadiazine are systemically absorbed. Therefore, it is possible that any adverse reactions associated with sulfonamides may occur.

Life-threatening cutaneous reactions (Steven-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN)) have been reported with the use of sulfadiazine. Patients should be advised of the signs and symptoms and monitored closely for skin reactions. The highest risk for occurrence of SJS or Ten is within the first weeks of treatment. If symptoms or signs of TEN (e.g. progressive skin rash often with blisters or mucosal lesions) are present, **Silverderm Cream** treatment should be discontinued. The best results in managing SJS and TEN come from early diagnosis and immediate discontinuation of any suspect drug. Early withdrawal is associated with a better prognosis. If the patient has developed SJS or TEN with the use of silver sulfadiazine, silver sulfadiazine must not be re-started in this patient at any time.

Use in hepatic/renal impairment

Silverderm cream should be used with caution in patients with impaired renal or hepatic function. Sensitivity has been shown to occur but the incidence is lower than with other sulfonamides.

Use in the elderly

No overall differences in safety or effectiveness were observed between these subjects and younger subjects, and other reported clinical experience has not identified differences in responses between the elderly and younger patients, but greater sensitivity of some older individuals cannot be ruled out.

Paediatric use

Silverderm cream should not be used in premature infants or newborn infants in the first months of life.

Effects on laboratory tests

In the treatment of burn wounds involving extensive areas of the body, the serum sulfonamide derivative concentrations may approach adult therapeutic levels (8mg % to 12mg %). Therefore, in these patients it would be advisable to monitor serum sulfonamide concentrations. Renal function should be carefully monitored and the urine should be checked for sulfonamide crystals. Absorption of the propylene glycol vehicle has been reported to affect serum osmolality, which may affect the interpretation of laboratory tests.

Drug Interactions

As the silver content of silver sulfadiazine may inactivate enzymatic debriding agents, their concomitant use may be inappropriate.

In large-area burns where serum sulfadiazine levels may approach therapeutic levels, it should be noted that the effects of systemically administered drugs may be altered. This can especially apply to oral hypoglycaemic agents and to phenytoin. In the case of these drugs, it is recommended that blood levels should be monitored as their effects can be potentiated.

Cimetidine: in patients with large area burns, it has been reported that coadministration of cimetidine may increase the incidence of leukopenia.

Sulfonamide may alter the effect of oral anticoagulant, methotrexate and cyclosporine. There are isolated reports that sulfonamide may also interfere with the effectiveness of hormonal contraceptive.

Use in pregnancy and lactation

Use in pregnancy- Pregnancy category C

Sulfonamides may cause kernicterus in babies during the first month of life by displacing bilirubin from plasma albumin. Sulfonamides should therefore be avoided as far as possible during the last month of pregnancy.

Use in lactation

Silverderm cream should be used with caution in breastfeeding mothers. Systemically, sulfadiazine can be excreted in breast milk although at concentration 15-35% of those found in serum.

Effect on Fertility

No data were available from studies in animal following topical administration of silver sulfadiazine.

Side Effects / Adverse Reactions

Blood & Lymphatic Tissue Disorders

Common: Leukopenia.

Leukopenia has been reported in 3-5% of burns patients treated with silver sulfadiazine. This may be a drug related effect, and often manifests itself 2-3 days after treatment has commenced. It is usually self-limiting and therapy with **Silverderm cream** does not usually need to be discontinued, although the blood count must be monitored to ensure that it returns to normal within a few days.

General Disorders & Administration Site Conditions

Common: Application site burning.

Renal & Urinary Disorders

Very rare: Renal failure.

Skin & Subcutaneous Tissue Disorders

Common: Pruritus, application site rash (including eczema and contact dermatitis).

Rare: Argyria.

There is evidence that in large area wounds and/or after prolonged application, systemic absorption of silver can occur causing clinical argyria.

Very rare: Severe cutaneous adverse reactions (SCARs), Steven-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN).

Immune system disorders

Unknown: Hypersensitivity including anaphylactic shock.

Symptoms and Treatment for Overdose

Not likely to occur with normal usage.

In extensively burned patients or in patients suspected of showing symptoms of excessive absorption, it is important to optimally maintain fluid balance, not only to prevent dehydration but also to avoid the possibility of renal failure.

If renal function is normal, fluids should be administered to maintain high urine output and assist in the rapid elimination of the drug.

Pharmacodynamics

Silver sulfadiazine is a sulfonamide and has broad antimicrobial activity against both Gram-positive and Gram-negative organisms.

Silver sulfadiazine acts on the cell membrane and cell wall. Unlike sulfadiazine or other sulfonamides, the antibacterial action of the silver salt of sulfadiazine does not appear to depend on inhibition of folic acid synthesis. Its action is not antagonised by p-aminobenzoic acid.

Silver sulfadiazine has broad antimicrobial activity against both Gram-positive and Gram-negative organisms including *Pseudomonas aeruginosa*, some yeasts and fungi. It has also been reported to be active in vitro against herpes virus and *Treponema pallidum*. Sulfonamides act by interfering with the synthesis of nucleic acids in sensitive micro-organisms by blocking the conversion of p-aminobenzoic acid to the co-enzyme dihydrofolic acid. Silver sulfadiazine has a bactericidal action; in contrast to sulfadiazine, the silver salt acts primarily on the cell membrane and cell wall and its action is not antagonised by p-aminobenzoic acid. Resistance to silver sulfadiazine has been reported and may develop during therapy.

Pharmacokinetics

Silver sulfadiazine is slowly metabolised in contact with wound exudates. Up to about 10% of the sulfadiazine may be absorbed; concentrations in blood of 10 to 20 µg/mL have been reported although higher concentrations may be achieved when extensive areas of the body are treated. Some silver may be absorbed.

There is evidence that in large area wounds and/or after prolonged application, systemic absorption of silver can occur causing clinical argyria. The sulfadiazine readily diffuses across wounds and enters the general circulation. The degree of uptake will significantly depend upon the nature of the wound and the dosing regimen. Sulfadiazine is excreted in the urine.

Packing / Pack Sizes

Aluminium tube of 25g and 50g.
Polyethylene jar of 450 g.

Storage Conditions, User Instructions & Pharmaceutical Precautions

Store below 30 °C in a well-closed container.
Protect from heat and direct light.
For external use only.
Keep out of reach of children.

Jauhi daripada kanak-kanak.

Shelf-life: Please refer to the outer box label.

Manufactured by:

T.O. Chemicals (1979) Ltd.

280 Soi Sabaijai, Suthisarnwinijai Road,
Samsennok, Huay-Kwang,
Bangkok 10310, Thailand.